One-Dimensional Discrete Random Variables

Probability Theory and Mathematical Statistics

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Chapter 2, Section 2

Outline

1. Definition and PMF
2. Representations of a PMF
3. PMF and CDF
4. Common Discrete Distributions
5. Poisson's Theorem
6. Summary and Exercises

Discrete Random Variables

Definition 2.1 (Discrete Random Variable)

A random variable X is **discrete** if its possible values are either finite or countably infinite:

$$X \in \{x_1, x_2, \ldots, x_n\} \quad \text{or} \quad X \in \{x_1, x_2, \ldots\}.$$

Definition 2.2 (Probability Mass Function, PMF)

For a discrete random variable X, the probabilities

$$p_k = P(X = x_k), \qquad k = 1, 2, \ldots$$

are called the **probability mass function** (PMF) of X.

Two Basic Properties of a PMF

Properties

1. **Nonnegativity:** $p_k \geq 0$ for all k;
2. **Normalization:** $\sum_k p_k = 1$.

Interpretation

The sum of the probabilities over all possible values of X equals 1.

Three Representations

1. **Table representation**

X	x_1	x_2	$\cdots$	x_k	$\cdots$
P	p_1	p_2	$\cdots$	p_k	$\cdots$

   or in matrix form

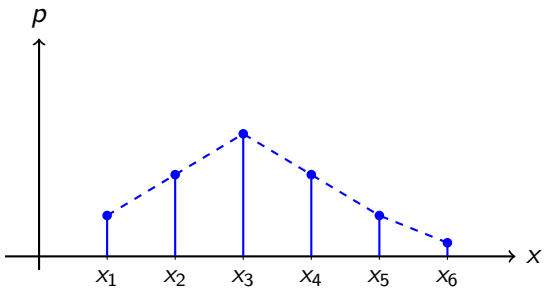
$$X \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_k & \cdots \\ p_1 & p_2 & \cdots & p_k & \cdots \end{pmatrix}.$$

2. **Formula representation**, e.g.,

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \qquad k = 0, 1, 2, \ldots$$

3. **Graphical representation**: vertical bars at (x_k, p_k).

Graphical Representation



Note

The heights of all bars sum to 1, i.e., $\sum_k p_k = 1$.

Proposition (PMF and CDF)

Proposition 2.1

Let X have PMF $p_k = P(X = x_k)$, $k = 1, 2, \ldots$. Then its CDF is

$$F(x) = P(X \leq x) = \sum_{x_k \leq x} p_k.$$

Moreover, $F(x)$ is a **right-continuous step function**:

- On each interval $[x_{k-1}, x_k)$, $F(x)$ is constant;
- At each possible value x_k, $F(x)$ has a jump of height p_k.

Conversely,

$$p_k = F(x_k) - F(x_{k-1}).$$

Example 2.1 (Statement)

Example 2.1

Let

$$P(X = 1) = \frac{1}{2}, \qquad P(X = 2) = \frac{1}{3}, \qquad P(X = 3) = \frac{1}{6}.$$

Find $F(x)$.

Example 2.1 (Solution)

Solution

- $x < 1$: $F(x) = 0$;
- $1 \le x < 2$: $F(x) = P(X = 1) = \frac{1}{2}$;
- $2 \le x < 3$: $F(x) = P(X = 1) + P(X = 2) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$;
- $x \ge 3$: $F(x) = 1$.

Hence

$$F(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{2}, & 1 \le x < 2, \\ \frac{5}{6}, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$

Remark

This CDF is a right-continuous step function with jumps of heights $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{6}$ at $x = 1, 2, 3$, respectively.

Degenerate Distribution

Definition

X takes the constant value c with probability 1:

$$P(X = c) = 1.$$

Denote $X \equiv c$.

PMF and CDF

$$P(X = c) = 1, \qquad F(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c. \end{cases}$$

Two-Point (Bernoulli) Distribution

Definition 2.3

X takes only 0 and 1, with

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p, \qquad 0 < p < 1.$$

Denote $X \sim B(1, p)$.

PMF

$$P(X = k) = p^k (1 - p)^{1-k}, \qquad k = 0, 1.$$

Background

Describes the number of "successes" in one Bernoulli trial, e.g., tossing a coin, inspecting a product.

Geometric Distribution

Definition 2.4

Let X be the number of trials needed for the **first success** in independent Bernoulli trials with success probability p. Then

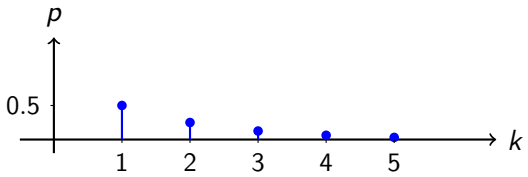
$$P(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, \ldots$$

Denote $X \sim G(p)$.

Normalization

$$\sum_{k=1}^{\infty}(1-p)^{k-1}p = p \cdot \frac{1}{1-(1-p)} = 1.$$

Geometric Distribution: Graph ($p = 1/2$)



Geometric Distribution: A Discrete Waiting Time

Remark (Geometric Distribution as a Discrete Waiting Time)

The geometric distribution describes a **discrete waiting time**: the system is in one of two states (success A or failure A^c) at each trial, and independent Bernoulli trials are repeated until the state changes (i.e., until the event A occurs for the first time). The number of trials X then follows the geometric distribution.

In other words:

- **Persist**: during the first $k-1$ trials, the system stays in the failure state A^c;
- **Stop at change**: at the k-th trial, the state changes to A for the first time, and the process stops.

Geometric Distribution: Parameter and Discrete–Continuous Pair

Meaning of the parameter

The parameter p can be interpreted as the **rate at which the event A occurs**: at each trial the probability of A is p; the larger p is, the faster the state changes.

Discrete–continuous pair

The geometric distribution forms a "discrete–continuous" pair with the **exponential distribution** introduced in Section 3:

- the geometric distribution describes the duration of a persistent state in discrete time;
- the exponential distribution describes the duration of a persistent state in continuous time (for example, the lifetime of a light bulb, from being switched on to failure).

Both have the **memoryless property**.

Proposition (Memoryless Property of the Geometric)

Statement

Let $X \sim G(p)$. Then for any positive integers $m, n \geq 1$,

$$P(X > m + n \mid X > m) = P(X > n).$$

Proof of the Memoryless Property

Proof

For $k \geq 1$,

$$P(X > k) = \sum_{j=k+1}^{\infty} (1-p)^{j-1} p = (1-p)^k.$$

By the definition of conditional probability,

$$P(X > m+n \mid X > m) = \frac{P(X > m+n)}{P(X > m)} = \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n = P(X > n).$$

Example 2.2 (Statement)

Example 2.2

In independent Bernoulli trials with success probability p, let

$$X_1 = \text{number of trials for the first success}, \quad X_2 = \text{number of trials for the second success}.$$

Find:

1. $P(X_1 = k,\ X_2 = n)$;
2. $P(X_1 = k \mid X_2 = n)$;
3. $P(X_2 = n \mid X_1 = k)$;
4. Explain the relation of these results to the memoryless property.

Example 2.2 (Solution, Part 1)

(1) Joint probability. For $1 \le k < n$:

$$P(X_1 = k,\ X_2 = n) = \underbrace{(1-p)^{k-1}p}_{\text{first success at } k} \cdot \underbrace{(1-p)^{n-1-k}}_{\text{failures in between}} \cdot \underbrace{p}_{\text{success at } n}.$$

Hence

$$P(X_1 = k,\ X_2 = n) = (1-p)^{n-2}p^2,$$

which is **independent of** k.

Example 2.2 (Solution, Part 2)

(2) Conditional probability $P(X_1 = k \mid X_2 = n)$.
First,

$$P(X_2 = n) = \binom{n-1}{1} p(1-p)^{n-2} \cdot p = (n-1)(1-p)^{n-2}p^2.$$

Therefore,

$$P(X_1 = k \mid X_2 = n) = \frac{(1-p)^{n-2}p^2}{(n-1)(1-p)^{n-2}p^2} = \frac{1}{n-1},$$

for $1 \leq k \leq n-1$.

Interpretation

Given that the second success occurs at trial n, the position of the first success is **uniformly distributed** on $\{1, 2, \ldots, n-1\}$.

Example 2.2 (Solution, Part 3)

(3) Conditional probability $P(X_2 = n \mid X_1 = k)$.
Since $X_1 \sim G(p)$,

$$P(X_1 = k) = (1-p)^{k-1}p.$$

Hence

$$P(X_2 = n \mid X_1 = k) = \frac{(1-p)^{n-2}p^2}{(1-p)^{k-1}p} = (1-p)^{n-k-1}p, \qquad k < n.$$

Interpretation

Given the first success at trial k, the additional number of trials $X_2 - k$ still follows $G(p)$. This is the **memoryless property**.

Example 2.2 (Solution, Part 4)

(4) Relation to the memoryless property.

The three results are:

1. $P(X_1 = k,\ X_2 = n)$ is independent of k;
2. $P(X_1 = k \mid X_2 = n) = \dfrac{1}{n-1}$: given that the second success occurs at trial n, the position of the first success is equally likely on $\{1, \ldots, n-1\}$;
3. $P(X_2 = n \mid X_1 = k) = (1-p)^{n-k-1}p$: given the first success at trial k, the **additional** number of trials $X_2 - k$ still follows $G(p)$.

All three results come from the same fact: the trials are independent, the success probability remains p, and past outcomes do not affect future outcomes. This is exactly the **memoryless property** of the geometric distribution:

$$P(X > m + n \mid X > m) = P(X > n), \qquad m, n \geq 1.$$

Binomial Distribution

Definition 2.5

Let X be the number of successes in n independent Bernoulli trials with success probability p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Denote $X \sim B(n, p)$.

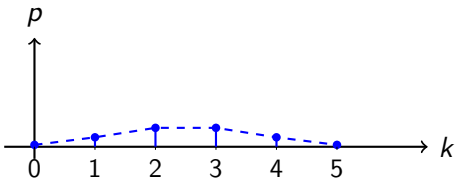
Normalization

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1.$$

Relation with the Two-Point Distribution

The two-point distribution is the special case $n = 1$: $B(1, p)$.

Binomial Distribution: Graph ($n = 5$, $p = 1/2$)



Example 2.3 (Statement)

Example 2.3

A shooter hits the target with probability 0.8. He fires 5 independent shots. Find the probability of exactly 3 hits.

Example 2.3 (Solution)

Solution

Let X be the number of hits. Then $X \sim B(5, 0.8)$, and

$$P(X = 3) = \binom{5}{3}(0.8)^3(0.2)^2 = 10 \times 0.512 \times 0.04 = 0.2048.$$

Poisson Distribution

Definition 2.6

A random variable X has a **Poisson distribution** with parameter $\lambda > 0$ if

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \qquad k = 0, 1, 2, \ldots$$

Denote $X \sim P(\lambda)$.

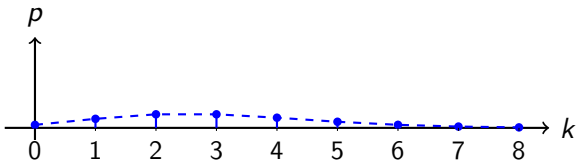
Normalization

$$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \cdot e^{\lambda} = 1.$$

Background

Models the number of rare events in a unit of time or space: radioactive decays, traffic accidents, telephone calls, etc.

Poisson Distribution: Graph ($\lambda = 3$)



Example 2.4 (Statement)

Example 2.4

The number of calls X per minute at a switchboard follows $P(3)$. Find:

1. $P(X = 2)$;
2. $P(X \geq 1)$;
3. $P(X \leq 2)$.

Example 2.4 (Solution)

Solution

$$P(X = k) = \frac{3^k}{k!}e^{-3}.$$

1. $P(X = 2) = \frac{3^2}{2!}e^{-3} = \frac{9}{2}e^{-3} \approx 0.2240$;
2. $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-3} \approx 0.9502$;
3. $P(X \leq 2) = e^{-3} + 3e^{-3} + \frac{9}{2}e^{-3} = \frac{17}{2}e^{-3} \approx 0.4232$.

Theorem (Poisson's Theorem)

Theorem 2.1 (Poisson)

Let $X_n \sim B(n, p_n)$, $n = 1, 2, \ldots$. If

$$np_n \longrightarrow \lambda > 0 \qquad (n \to \infty),$$

then for every fixed nonnegative integer k,

$$\lim_{n\to\infty} P(X_n = k) = \lim_{n\to\infty} \binom{n}{k} p_n^k (1 - p_n)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}.$$

Proof of Poisson's Theorem (1)

Let $\lambda_n = np_n$. Since $np_n \to \lambda$, we have

$$\lambda_n \to \lambda, \qquad p_n = \frac{\lambda_n}{n} \to 0.$$

For fixed k,

$$P(X_n = k) = \binom{n}{k} p_n^k (1-p_n)^{n-k} = \frac{\lambda_n^k}{k!} \cdot \underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{A_n} \cdot \underbrace{\left(1 - \frac{\lambda_n}{n}\right)^n}_{B_n} \cdot \underbrace{\left(1 - \frac{\lambda_n}{n}\right)^{-k}}_{C_n}.$$

Proof of Poisson's Theorem (2)

Limits of the three factors

1. $A_n \to 1$: since k is fixed,
$$A_n = \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right) \to 1.$$
2. $B_n \to e^{-\lambda}$: by the standard limit
$$B_n = \left(1 - \frac{\lambda_n}{n}\right)^n \to e^{-\lambda}.$$
3. $C_n \to 1$: since $\lambda_n/n \to 0$,
$$C_n = \left(1 - \frac{\lambda_n}{n}\right)^{-k} \to 1.$$

Proof of Poisson's Theorem (3)

Since $\lambda_n^k \to \lambda^k$, combining the limits gives

$$\lim_{n\to\infty} P(X_n = k) = \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda} \cdot 1 = \frac{\lambda^k}{k!} e^{-\lambda}.$$

Hence the limiting distribution of X_n is $P(\lambda)$. □

Remark (Conditions for Poisson Approximation)

Remark

When n is large, p is small, and $np = \lambda$ stays moderate, the binomial distribution $B(n, p)$ can be approximated by the Poisson distribution $P(\lambda)$:

$$\binom{n}{k} p^k (1-p)^{n-k} \approx \frac{\lambda^k}{k!} e^{-\lambda}, \qquad \lambda = np.$$

In practice, the approximation is already quite accurate when

$$n \geq 50, \qquad p \leq 0.1.$$

The larger n is and the smaller p is, the better the approximation.

Example 2.5 (Statement)

Example 2.5

Let $n = 100$, $p = 0.02$, so $\lambda = np = 2$. Compare $B(100, 0.02)$ with $P(2)$ for $k = 0, 1, 2, 3$.

Example 2.5 (Numerical Illustration)

k	Binomial $B(100, 0.02)$	Poisson $P(2)$
0	$\binom{100}{0}(0.02)^0(0.98)^{100} \approx 0.1326$	$\frac{2^0}{0!}e^{-2} \approx 0.1353$
1	$\binom{100}{1}(0.02)^1(0.98)^{99} \approx 0.2707$	$\frac{2^1}{1!}e^{-2} \approx 0.2707$
2	$\binom{100}{2}(0.02)^2(0.98)^{98} \approx 0.2734$	$\frac{2^2}{2!}e^{-2} \approx 0.2707$
3	$\binom{100}{3}(0.02)^3(0.98)^{97} \approx 0.1823$	$\frac{2^3}{3!}e^{-2} \approx 0.1804$

Even for moderate $n = 100$, the difference is already very small.

Example 2.6 (Statement)

Example 2.6

A factory produces items with defect rate $p = 0.01$. Randomly select $n = 200$ items. Find the probability of exactly 2 defective items.

Example 2.6 (Solution)

Solution

Let $X \sim B(200, 0.01)$. Since n is large, p is small, and

$$\lambda = np = 200 \times 0.01 = 2,$$

we use $P(2)$:

$$P(X = 2) \approx \frac{2^2}{2!} e^{-2} = 2e^{-2} \approx 0.2707.$$

The exact binomial value is

$$P(X = 2) = \binom{200}{2} (0.01)^2 (0.99)^{198} \approx 0.2720.$$

Remark

The Poisson approximation is already quite accurate when $n \geq 50$ and $p \leq 0.1$. The larger n is and the smaller p is, the better the approximation.

Summary: Basic Concepts and Representations

Key points

1. **Discrete random variable**: possible values are finite or countably infinite.
2. **PMF**: $p_k = P(X = x_k)$, with $p_k \geq 0$ and $\sum_k p_k = 1$.
3. **Three representations**: table, formula, graph.
4. **Relation with CDF**: $F(x) = \sum_{x_k \leq x} p_k$, a right-continuous step function with jumps of height p_k.

Summary: Common Distributions and Important Properties

Common distributions

- Degenerate: $P(X = c) = 1$;
- Two-point (Bernoulli): $B(1, p)$;
- Geometric: $G(p)$, $P(X = k) = (1 - p)^{k-1}p$;
- Binomial: $B(n, p)$, $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$;
- Poisson: $P(\lambda)$, $P(X = k) = \dfrac{\lambda^k}{k!} e^{-\lambda}$.

Summary: Common Distributions and Important Properties

Important properties

- Poisson's theorem: when n is large, p is small, and $np = \lambda$,

$$\binom{n}{k} p^k (1-p)^{n-k} \approx \frac{\lambda^k}{k!} e^{-\lambda}.$$

- Memoryless property of the geometric distribution:

$$P(X > m + n \mid X > m) = P(X > n).$$

Exercises (Part 1)

1. **(Normalization)** Let

$$P(X = k) = c \cdot \frac{1}{2^k}, \qquad k = 1, 2, 3, \ldots$$

   Find c.

2. **(CDF and probabilities)** Let

$$F(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \le x < 0, \\ 0.5, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

   Find $P(X = 0)$, $P(X < 0)$, $P(0 < X \le 1)$.

3. **(Geometric)** A shooter hits with probability $p = 0.4$. Find the probability that he hits for the first time on the 3rd shot, and the probability that he hits within the first 3 shots.

Exercises (Part 2)

4. **(Binomial)** Defect rate $p = 0.02$, sample $n = 20$. Find the probability of exactly 1 defective item and the probability of at least 1 defective item.
5. **(Additivity of Binomial)** Two machines independently produce items with defect rate $p = 0.05$. Sample 8 items from machine A and 12 from machine B. Find the probability of exactly 2 defective items in the combined sample.
6. **(Poisson)** Calls arrive at rate $\lambda = 4$ per minute. Find the probability of exactly 2 calls and the probability of at most 1 call in a minute.

Exercises (Part 3)

7. **(Negative Binomial)** A shooter hits with probability $p = 0.3$. He fires independently until the 3rd hit. Find:
   1. the probability that he needs exactly 8 shots to achieve the 3rd hit;
   2. the probability that he needs at most 5 shots to achieve the 3rd hit.
8. **(Negative Binomial and Geometric)** In independent Bernoulli trials with success probability p, let X be the number of trials needed for the r-th success.
   1. Write down the PMF of X;
   2. State which distribution X follows when $r = 1$;
   3. Show that X can be represented as the sum of r independent geometric random variables.

Thank you!